Candidate Seat no…………

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Eighth Semester of B. Tech. (CL) Examination

April-May 2018

CL 411 Structural Dynamics & Earthquake Engineering

Date: 30.04.2018, Monday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures with pencil only wherever required.
4. Use of scientific calculator is allowed.
5. IS 1893:2016 Part-I, IS 4326:1993, IS 13920:2016, IS 456:2000 are permitted, provided that they do not contain anything other than the printed matter inside.
6. Assume Fe 415 and M 20 unless otherwise mentioned.

SECTION – I

Q-1 Choose the best alternative in the following: **[07]**

a) Which type of vibrations are also known as transient vibrations?
(i) Damped vibrations (ii) Undamped vibrations
(iii) Torsional vibrations (iv) Transverse vibrations

b) The product of frequency and time period is equal to
(i) 1 (ii) 2
(iii) ½ (iv) 0.5π

c) The energy absorbed by a body is _____________ at resonance.
(i) Minimum only (ii) Maximum only
(iii) Maximum as well as minimum (iv) Zero

d) What is the maximum distance of first shear reinforcement bar from the edge of column in a beam?
(i) 40 mm (ii) 35 mm
(iii) 50 mm (iv) 30 mm

e) For flexural member, what is the preferably width to depth ratio
(i) More than 0.3 (ii) More than 0.2
(iii) Less than 0.3 (iv) None of these

f) What should be minimum diameter of the bar forming a hoop (link)?
(i) 6 mm (ii) 8 mm
(iii) 12 mm (iv) 10 mm

g) As per IS 13920:2016, lap length shall not be less than the ____________ of the largest longitudinal reinforcement bar in tension.
(i) Development length (ii) Quarter length of the beam
(iii) Effective length of the beam (iv) None of these

Q-2 Attempt any TWO: **[14]**

1) Derive the equation of displacement for the force undamped vibration of SDOF system.

2) A water tank is idealized as a single degree of freedom having equivalent weight of 10^5 N, damping ratio as 4% and stiffness factor as 2×10^6 N/m. Calculate the maximum horizontal displacement at the top of the water tank if it is loaded by a seismic force equivalent to 20sin(5t) N.

3) Explain logarithmic decrement. A SDOF system having the amplitude of vibration in successive cycle are 1.520, 0.950, 0.594, 0.371 units respectively. Determine damping ratio of the system.

Q-3(A) A fixed-ended RC beam of rectangular section 300 mm × 600 mm having five number of 20 mm diameter bars of bottom and two number of 12 mm diameter bars of top. Centre-to-centre distance between supports is 6 m. Draw and detail the beam satisfying all criteria of IS 13920:2016. **[04]**

Q-3(B) Attempt any ONE: **[10]**

1) Define transmissibility and its importance. A machine of weight W = 17370 N is mounted on a 3.05 m simply supported steel beam at center. A piston that moves up and down in the machine produces a harmonic force of magnitude F_0 = 31500 N and frequency ω = 60 rad/sec. Neglecting the weight of the beam and assuming 10% of the critical damping, determine the amplitude of the motion of the machine. Take $E = 2.09 \times 10^{11}$ N/m², $I = 5 \times 10^{-5}$ m⁴.

2) Define dynamic load. Enlist various type of dynamic load acting on a structure.
For the two storey building frame having lumped masses 10 tons at both floor levels having storey stiffness 50 kN/m on both level. Calculate natural frequencies & draw all mode shapes. Also show that mode shapes are orthogonal.

SECTION – II

Q-4(A) Choose the best alternative in the following: **[04]**

a) The storey drift in any storey due to the minimum specified design lateral force, with partial safety factor 1.0 shall not exceed _________ time the storey height.
(i) 0.004 **(ii)** 0.4
(iii) 0.0004 **(iv)** None of these

b) Point at which the resultant of restoring forces of systems acts is _______________.
(i) Center of mass **(ii)** Center of stiffness
(iii) At mid height of column **(iv)** None of these

c) Response reduction factor for ductile shear wall with special moment resisting frame is
(i) 3 **(ii)** 2
(iii) 4 **(iv)** 5

d) The first longest modal time period of vibration is called _______________.
(i) Fundamental natural time period **(ii)** Natural period
(iii) Normal mode **(iv)** Fundamental normal mode

Q-4(B) State whether following statements are true or false & explain reasons for the same in brief: **[03]**

1) Code specifies higher value of R for building having better performance.

2) Can my building withstand a magnitude 8.0 earthquake?

3) Strong beam-weak column concept is preferred in earthquakes resistant design.

Q-5 (A) Explain the concept of ductile-brittle chain behavior. **[02]**

(B) Draw neat sketch of stirrups detailing in beam as per IS 13920:2016. **[02]**

(C) **Attempt any TWO:** **[10]**

1) To ensure good seismic performance of masonry building, all walls of masonry building must be joined properly to the adjacent walls. Justify this statement with suitable sketches.
2) Define the bands used in building. At which levels in a masonry building the bands are provided? Give justifications for each of them.
3) **Figure 1** shows the plan of a single storey brick masonry ordinary residential building located in Vadodara. Consider all walls of 230 mm thickness. Decide the position of door (D), window (W) and maximum size of window that can be provided in wall 'A', 'B', and 'C' considering width of door as 900 mm.

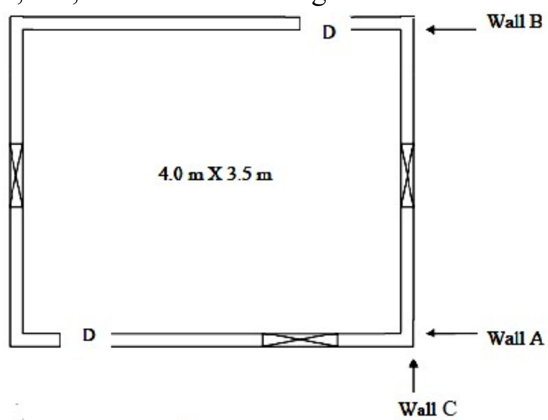

Figure 1

Q-6(A) For the building shown in **Figure 2**, locate the center of mass. **[03]**

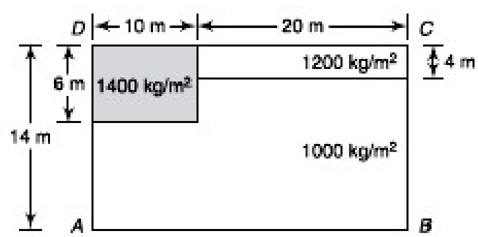

Figure 2

(B) Explain seismic design philosophy of earthquake resisting structures. **[04]**

(C) **Figure 3** shows the plan of a G + 1 storied building with storey height of 3.6 m, all beams of 250 x 500 mm and all columns of size 300 x 600 mm. Assume 230 mm thick full height brick masonry wall around outer periphery of building at all level, slab of 130 mm thickness, live load = 5 kN/m^2 and floor finish = 1 kN/m^2. Consider plinth height as 1.2 m from footing level. Calculate seismic weight and center of mass of Level 2 in X-direction. **[07]**

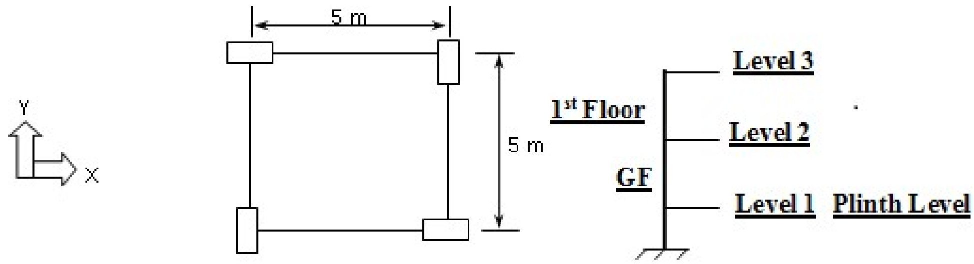

Figure 3

OR

Q-6(A) Determine the lateral forces on a two-storey unreinforced brick masonry building, as shown in **Figure 4**, situated near Allahabad (zone III) for the following data: **[06]**

Plan size =18 m × 8 m
Total height of the building = 6.2 m
Storey height = 3.1 m
Weight of roof = 2.5 kN/m²
Weight of wall = 5.0 kN/m²
Live load on roof = 0
Live load at floor = 1.0 kN/m²
Soil: (Type II) medium soil

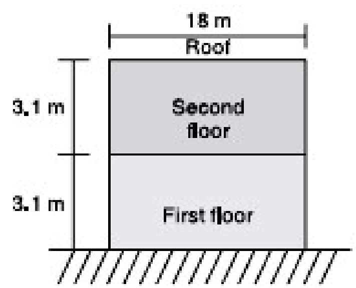

Figure 4

(B) Enlist modal combination methods used to find response of different modes considered for dynamic analysis. **[08]**

A three storey school building has lumped floor mass from bottom to top as W_1 = 640 kN, W_2 = 688 kN, and W_3 = 688 kN. The building is located in seismic zone V. The type of soil encountered is medium stiff and it is proposed to design the building with a special moment resisting frame. Determine the design seismic forces for the building using dynamic analysis. Use any of the modal combination method to find storey shear forces. Given the vibration analysis results as follows:

Frequency $\{\omega\} = \{11.79, 32.93, 47.04\}$ rad/sec

Modes: $\{Ø_1\} = \{0.45, 0.81, 1.000\}$

$\{Ø_2\} = \{-1.223, -0.489, 1.000\}$

$\{Ø_3\} = \{1.611, -2.038, 1.000\}$

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